

2 *Use of the Data Booklet is relevant to this question.*

W is an isotope of an element. An isotope of strontium has a nucleon number of 84 and the same number of neutrons in the nucleus as the isotope **W**.

The species Sr^{2+} and **W** contain the same number of electrons.

What is the nucleon number of **W** and how many orbitals are occupied in its valence shell?

	W	number of orbitals occupied in valence shell
A	80	3
B	82	4
C	82	3
D	84	4